

NORTHOLT, United Kingdom

WMO: 036720

Lat: 51.5486N

Lon: 0.4169W

Elev: 108

StdP: 14.64

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.2	27.7	19.7	14.8	31.9	22.4	16.9	32.9	28.9	50.2	25.4	49.4	3.4	330	0.566

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.9	83.1	65.5	79.2	63.9	75.8	62.6	67.5	78.8	65.6	75.3	64.1	72.7	8.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.4	87.9	70.6	61.9	83.2	69.2	60.6	79.3	67.8	31.9	78.5	30.5	75.1	29.3	72.9	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.0	20.4	18.2	19.6	89.5	3.5	4.0	17.1	92.4	15.1	94.7	13.1	97.0	10.6	99.9		
(5)			18.9	71.6	3.4	2.4	16.4	73.3	14.4	74.7	12.5	76.0	10.0	77.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.0	41.0	41.8	45.1	49.6	55.2	60.8	64.8	64.2	59.2	53.2	45.9	41.9		
	DBStd	10.15	6.15	5.84	5.45	5.50	5.52	4.83	4.88	4.70	4.53	5.40	5.89	6.72		
	HDD50	1208	283	234	166	71	15	0	0	0	0	30	147	262		
	HDD65	4894	745	650	617	461	305	143	64	71	182	367	572	717		
	CDD50	1924	3	4	15	60	177	325	460	441	276	129	25	9		
	CDD65	134	0	0	0	0	3	17	59	47	7	1	0	0		
	CDH74	1150	0	0	0	6	44	187	483	363	62	5	0	0		
	CDH80	274	0	0	0	0	2	41	125	98	7	1	0	0		
(14) Wind	WSAvg	8.5	9.3	9.6	9.3	8.5	8.9	8.5	8.1	7.5	7.7	8.1	7.9	8.7		
(15) Precipitation	PrecAvg	23.7	2.1	1.4	1.7	1.9	1.9	2.0	1.7	2.0	2.1	2.4	2.3	2.2		
	PrecMax	32.3	4.5	3.6	3.7	4.0	4.0	5.0	3.7	5.9	5.5	6.9	5.9	4.7		
	PrecMin	16.6	0.4	0.1	0.2	0.2	0.2	0.2	0.3	0.0	0.1	0.1	0.3	0.4		
	PrecStd	3.9	1.1	0.8	1.0	0.9	1.0	1.2	0.9	1.2	1.5	1.5	1.4	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.8	58.8	64.9	74.0	78.7	85.2	88.3	87.9	80.3	71.5	61.1	57.6		
		MCWB	52.2	51.1	51.8	57.8	61.8	66.8	67.9	67.6	64.4	62.1	57.0	53.7		
	2%	DB	54.1	54.9	59.9	67.4	74.4	79.2	83.4	82.3	74.9	66.0	58.8	55.6		
		MCWB	50.9	50.0	50.4	54.8	60.5	63.8	65.2	65.7	62.1	59.1	54.9	52.4		
	5%	DB	52.2	52.7	56.9	63.1	70.2	74.9	79.4	77.7	70.8	63.5	57.0	54.0		
		MCWB	49.1	48.5	49.7	52.5	58.4	61.7	64.1	63.9	60.6	57.9	53.6	51.1		
	10%	DB	50.4	50.8	54.2	59.7	66.1	71.3	75.5	73.9	67.9	61.5	55.1	52.1		
		MCWB	47.4	46.9	48.3	50.9	56.1	59.8	62.7	62.3	59.0	56.8	51.7	49.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.3	52.9	54.2	59.8	64.8	68.3	69.8	69.6	66.8	63.1	58.1	54.7		
		MCDB	55.1	56.3	60.1	70.8	74.7	82.2	83.8	84.4	75.0	69.2	60.1	56.3		
	2%	WB	51.5	50.7	52.2	56.1	61.7	65.2	67.4	67.3	64.3	60.7	55.8	53.2		
		MCDB	53.7	54.0	57.1	65.4	70.7	75.5	79.0	78.2	71.6	64.3	58.0	55.2		
	5%	WB	49.5	49.1	50.8	53.8	59.4	62.9	65.5	65.4	62.4	59.1	54.0	51.5		
		MCDB	51.9	52.1	55.4	60.9	67.9	72.1	75.8	74.2	68.0	62.2	56.5	53.8		
	10%	WB	47.8	47.5	49.1	51.8	57.4	61.2	63.8	63.8	60.8	57.7	52.0	49.5		
		MCDB	50.1	50.4	53.6	58.4	64.9	69.5	73.2	71.5	65.9	60.9	54.6	51.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.2	11.4	13.8	16.3	16.4	16.7	16.9	16.7	16.1	13.6	11.5	9.9		
		MCDBR	10.4	13.3	18.5	22.6	23.2	23.6	24.1	23.1	20.9	15.2	11.1	9.4		
		MCWBR	9.6	9.7	11.1	12.2	11.6	11.2	9.6	9.6	10.4	9.6	9.0	8.6		
	5% WB	MCDBR	10.3	12.2	15.6	19.9	20.2	20.4	20.4	19.6	17.1	13.3	11.0	9.6		
		MCWBR	9.9	9.9	10.5	11.4	10.8	10.8	9.4	9.3	9.8	9.7	9.7	9.2		
(40) Clear-Sky Solar Irradiance	taub		0.331	0.347	0.384	0.405	0.409	0.410	0.405	0.396	0.378	0.361	0.334	0.320		
	taud		2.365	2.370	2.262	2.220	2.249	2.275	2.295	2.340	2.388	2.417	2.410	2.350		
	Ebn at Noon		213	244	255	263	268	268	267	264	255	236	211	196		
	Edh at Noon		20	26	35	40	41	40	39	36	30	24	19	17		
(44) All-Sky Solar Radiation	RadAvg		257	465	808	1248	1512	1671	1584	1316	993	577	306	202		
	RadStd		22	62	110	144	97	168	157	107	85	52	35	24		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (83 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page